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Ai-enabled framework for anomaly detection in power distribution networks under false data injection attacks

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Abstract

Modern power distribution networks are becoming cyber physical systems due to the addition of more advanced metering infrastructure (AMI). This has introduced new vulnerabilities to cyber threats, particularly false data injection (FDI) attacks. These attacks compromise the integrity of power consumption data, leading to financial losses, operational inefficiencies, and grid instability. Rule-based techniques and traditional machine learning models are two examples of traditional anomaly detection methods that often have problems. Often, these methods generate an excessive number of false alarms, struggle to adapt to new attack patterns, and perform poorly in large-scale deployments. This research suggests a robust anomaly identification framework (AIF) that uses an autoencoder (AE) for feature transformation and a multi-layer perceptron (MLP) to identify anomalies in AMI integrated with smart grids. The proposed approach first applies synthetic features extraction inspired by real-world smart meter capabilities and transforms the dataset using a denoising AE. MLP assisted in the classification to detect multiple FDI attack types with improved accuracy and reliability. Numerous experiments have been performed, and the results indicate that the suggested method works better than popular methods like correlation analysis, techniques based on clustering, and standard outlier identification algorithms. Compared to baseline methods, the proposed technique improves detection accuracy by up to approximately 25%, reduces false positives, and enhances the system's ability to generalize across different cyberattack strategies. The proposed work computes seven different types of criterion matrices to verify the effectiveness of finding anomalies. The overall average results include mean squared error (0.0793), accuracy (92%), F1-Score (92%), recall (91%), specificity (94%), area under the curve (97%), and mean average precision (96%). These findings accentuate the potential of the proposed AIF performance in fortifying smart grid cybersecurity.

Keywords Cyber physical system · Modern power systems · False data injections · Machine learning · Anomaly identification framework

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1 Introduction

1.1 Background

Modern energy transmission and distribution power networks face increasing vulnerability to non-technical losses, such as false data injection (FDI) attacks and other deceptive practices. Multiple studies have estimated that non-technical losses in the global energy sector amount to approximately 89.3 billion USD annually (De Souza Savian et al. 2021; Li et al. 2022; Irfan et al. 2025; Mamoon et al. 2024). To address the issue of manipulated energy consumption data, many countries have adopted advanced monitoring systems within their power infrastructures (Yang et al. 2025; Zhang et al. 2025). These systems collectively referred to as advanced metering infrastructure (AMI), that is integration of smart meters, communication technologies, and information technology (IT) systems. This addition transforms the power distribution system into a smart grid (SG), a representation of a contemporary cyber physical system (CPS), and is a crucial element in the effort to combat cyber threats in advanced power systems (Mamoon et al. 2024; Zhou et al. 2025).

The Fig. 1 represents the CPS, which integrates physical processes with computational and communication technologies. Among the essential components of CPS, sensors are used to collect real-time data; communication systems enable secure data exchange; actuators are responsible for executing control actions; Storage devices manage and archive data; and computing hardware processes and analyzes information to make intelligent decisions (Zhang et al. 2024). The interconnected nature of these components ensures seamless interaction between the cyber and physical domains. These components makes CPS vital for applications in SGs, industrial automation, healthcare, and autonomous systems (Mathumitha et al. 2024; Ahmad et al. 2025; Habib et al. 2025).

Smart meters are classified as industrial grade devices with internet-of-things (IoT) enabled characteristics. They are vulnerable to various forms of cyberattacks and malicious tampering attempts (Anjum et al. 2025). There are multiple possibilities that the unauthorized actors may alter legitimate meter readings and physically damage the device (Liu et al.

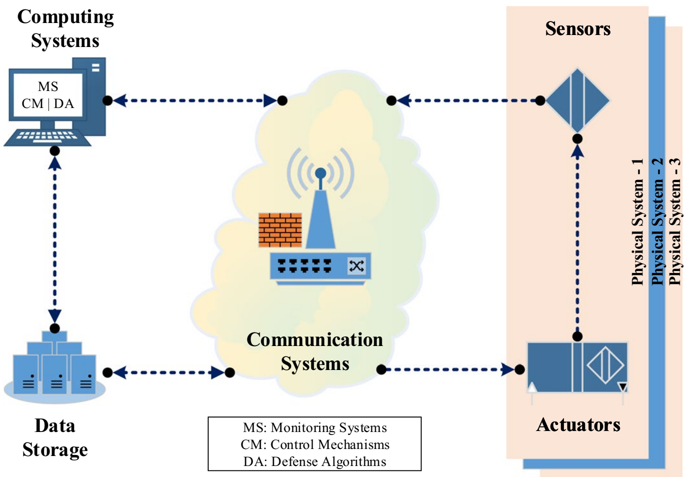


Fig. 1 A conceptional view of a modern CPS with generic components (Guo et al. 2024)

2022; Thakur et al. 2023; Ahmed et al. 2024). Different operational scenarios emphasized the complexity and diversity of non-technical energy losses. Numerous types of FDI attacks with each exhibiting unique characteristics, have been thoroughly investigated in the existing literature (Giraldo et al. 2022; Mohammadi and Rashidzadeh 2022; Ahmad et al. 2024; Cali et al. 2024). The SG-based power system architecture and its associated vulnerable components are depicted in Fig. 2.

1.2 Problem statement

IoTs technology advancements in the power sector, especially the deployment of AMI, have enhanced SG systems' ability to identify anomalies by analyzing high-resolution energy consumption data. The introduction of smart meters marks a noteworthy development in AMI, that enables the collection of detailed electrical usage data at frequent and desired intervals. This data supports accurate modeling of consumer usage behavior (Zjavka 2023; Liaqat et al. 2020). That helps in improving demand forecasting and price estimation (Ahmed et al. 2022; Habib et al. 2024) and facilitates the execution of demand side management strategies (Khajeh and Laaksonen 2022; Khan et al. 2021).

Guaranteeing the secure data transmission in IoT-enabled electrical networks remains a significant challenge. That is particularly due to vulnerabilities associated with smart meters. As illustrated in Fig. 3, AMIs subjected to threats are categorized into three main types. These includes physical, software, and communication breaches. Physical tampering involves techniques such as using strong magnets to bypass measurements, damaging internal components, altering legitimate hardware, exploiting optical interfaces, and physically inverting the meter. Each of these attempts are capable of impairing accurate meter operation of smart meters. Software-based attack vectors include unauthorized access through BIOS manipulation, software modification, resource exhaustion, deletion of logs, and disabling meter functions. These vectors raise critical concerns about the reliability and functionality of SG control and power delivery systems. Communication related vulnerabilities further endanger the AMI, where weak data encryption algorithms and communication

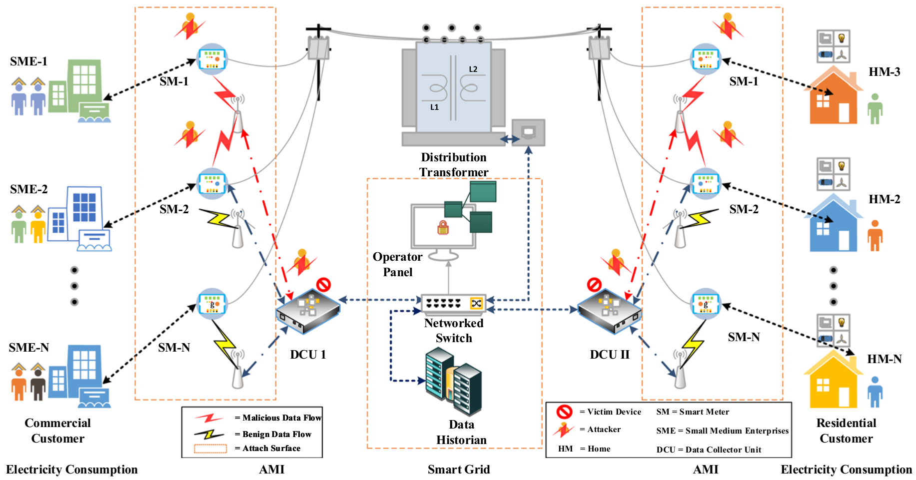


Fig. 2 Smart grid enabled with AMI for power distribution network

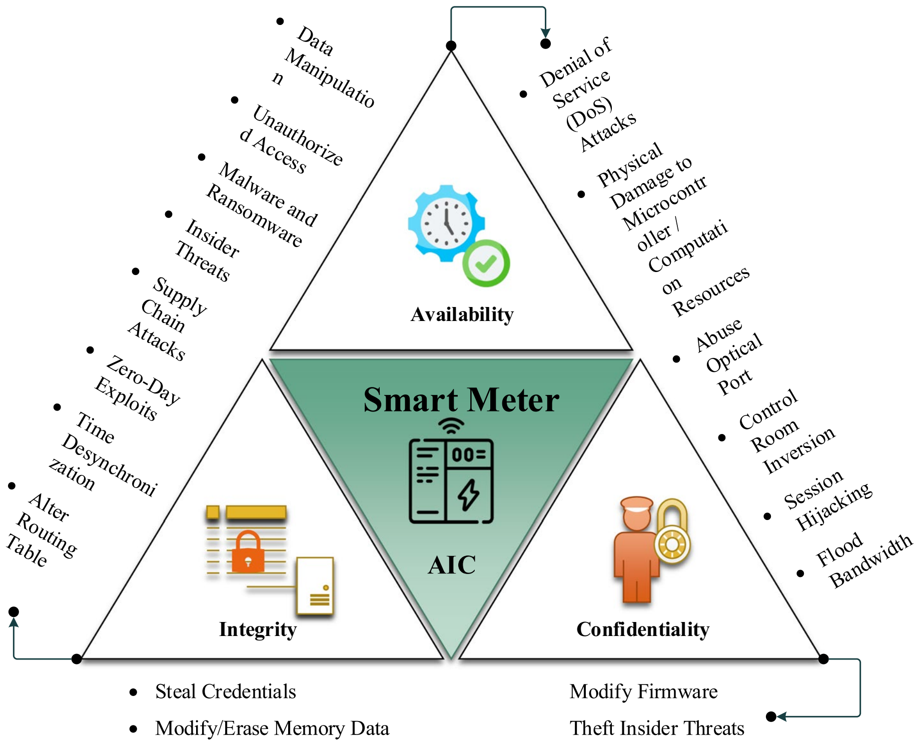


Fig. 3 Type of breaches possible on smart meter (Ahmad et al. 2025)

flaws can be exploited to intercept data and/or gain unauthorized system access (Yang et al. 2022; Ganjkhani et al. 2021; Ahmed et al. 2023). Cyberattacks such as man-in-the-middle, packet drops, session hijacking, spoofing, and denial-of-service can severely interrupt real-time monitoring and the stability of AMI. Therefore, it is essential to develop and implement robust data-driven/analytical methods to detect anomalies in power distribution infrastructure (Sana et al. 2023; Rahman et al. 2022; Khalid et al. 2024).

1.3 Related works

Although AMI has provided considerable benefits to SGs, it has also introduced several new security and safety vulnerabilities. Conventionally, threats involve physical tampering, such as removing and/or damaging mechanical meters to reduce energy consumption bills (Haq et al. 2023; Basit et al. 2024). However, with the implementation of AMI, new prospects have emerged for malicious intruders to commit energy theft using advanced digital methods. These include manipulating data produced by smart meters and carrying out targeted cyber intrusions. As stated in a recent research study, Mexico's federal electricity commission experiences an estimated annual loss of approximately 2.5 billion USD due to FDIs in energy consumption records (Qi et al. 2022). Traditional methods for detecting these intrusions typically relies on manual inspections and visual surveillances. These are time consuming and labor intensive. With the availability of detailed data from AMI, modern data-driven techniques have been developed to overcome the shortcomings of these conventional practices. As outlined in (Gu et al. 2022), data analysis-based methods for identifying FDIs can be classified into two main categories, i.e. supervised and unsupervised learning.

State estimation-based methods are utilized to assess the condition of the energy system in order to identify irregularities in energy consumption data. In (Sethi et al. 2022), the authors proposed a game-theoretic framework to model the adversarial nature of falsified data within AMI environments. However, a challenge associated with this class of methods is defining a suitable function that accurately captures the interaction between utility providers and consumers.

In (Khan et al. 2025), the authors proposed a machine learning framework that combines boosting algorithms such as adaptive boost and gradient boosting decision trees to detect falsified energy consumption data within AMI. The advantage of their approach lies in its ability to enhance detection accuracy through ensemble learning. That effectively handles data imbalance and captures complex theft patterns. The data-driven nature of the framework enables scalable deployment across different consumer categories. However, the primary limitation of this study is its reliance on supervised learning, which necessitates labeled data for both normal and malicious consumption patterns. As an aspect that may not always be feasible in real-world power distribution environments. Authors in (Qi et al. 2022) investigated and developed an unsupervised method that utilizes observer meters to detect discrepancies in user consumption patterns, that offers a practical solution for scenarios with unavailability of labeled data. Authors have utilized discrete wavelet transform for removal of outliers and fuzzy c-means algorithm for clustering of normal and benign energy consumer. This makes it suitable for real-world AMI applications. However, its effectiveness depends heavily on the correct placement and reliability of observer meters. It may struggle to detect sophisticated FDI attacks that closely mimic normal consumption behavior. Similarly (Mishra and Das 2025), proposed a hierarchical density-based spatial clustering of applications with noise-based clustering method to distinguish between benign and abnormal consumption patterns of AMI collected data. However, its performance may sensitive to parameter selection such as neighborhood radius. This may struggle with overlapping clusters and refined FDI attacks that do not produce clear outliers.

(Wang et al. 2025) introduced a multi-scale feature fusion transformer detector that uses semi-supervised learning to compensate limited labeled data by capturing long-range dependencies and fusing multi-scale features. Similarly in (Shahzadi et al. 2024) focused on urbanized smart grids. They employed AI to identify and mitigate energy theft in densely populated areas. Authors have combined clustering and local outlier factor (LOF) to detect anomalous consumption (Peng et al. 2021), that enhanced detection accuracy by identifying deviations from typical consumer usage patterns. (Zheng et al. 2018) developed a combined data-driven framework, that integrates multiple analysis techniques to improve the robustness of anomalies detection systems.

1.4 Research gap and outcomes

In recent times, the utilization of machine learning methods has significantly increased for detecting anomalies (Chen et al. 2024) and uncovering scenarios of manipulated energy consumption by analyzing consumer usage patterns. The aforementioned techniques predominantly employ computations involving unsupervised as well as supervised learning.

The technique of supervised learning involves the utilization of labeled information that comprises load patterns that include legitimate and deceitful clients, with the aim of constructing identification models. A variety of supervised learning algorithms have been used to develop classification models for detecting falsified energy consumption data, including support vector machines and artificial neural networks (Avila et al. 2018). The issue was addressed by utilizing deep learning designs, including convolutional neural networks, also called CNNs (Zheng et al. 2017). Given that the majority of individuals in real-life situations are typical as opposed to forged, it is possible for the information utilized in supervised learning to exhibit a significant degree of asymmetry. A framework constructed through unsupervised training may encounter the issue of excessive fitting, thereby impeding its ability to identify novel forms of attacks and process information from previously unseen clients.

A comparison among various methods from scholarly works and the proposed method is presented in Table 1. The proposed anomaly identification framework (AIF) offers several noteworthy features compared to existing methodologies. Unlike the approaches investigated in Mishra and Das (2025); Wang et al. (2025); Shahzadi et al. (2024); Peng et al. (2021); Zheng et al. (2018), the proposed method integrates statistical correlation, synthetic feature creation, and feature transformation, that ensure a comprehensive feature extraction process. Compared to Qi et al. (2022); Gu et al. (2022), which lacks a scalability, the proposed AIF integrated with a hybrid machine learning approach included with auto-encoder and multilayer perceptron (MLP), that increases it scalability and make it more robust for detecting multiple type of FDIs. This significantly improves the adaptability and efficiency in handling large-scale power distribution networks integrated with smart grids. While existing methods such as Mishra and Das (2025); Wang et al. (2025); Peng et al. (2021); Zheng et al. (2018); Sethi et al. (2022) fail to cover all FDI attack types, the proposed investigates seven FDI attack types, that ensures a more robust anomaly detection mechanism. Moreover, in contrast to Mishra and Das (2025); Wang et al. (2025); Shahzadi et al. (2024); Peng et al. (2021); Zheng et al. (2018); Gu et al. (2022); Sethi et al. (2022), that do not perform an ablation study, the proposed systematically evaluates each component's contribution. This ensures enhancement in the reliability of the results. The integration of these advanced features demonstrates the superiority of the proposed AIF, that makes it a more effective and scalable solution for anomaly detections in power distribution networks under multiple cyberattacks.

Table 1 Feature comparison among proposed and scholarly works

References	Statistical correlation	Synthetic features creation	Feature transformation	Hybrid machine learning	Scalable AIF	FDI types	Ablation study
Mishra and Das (2025)	×	×	×	×	×	5	×
Wang et al. (2025)	×	×	×	×	×	6	×
Shahzadi et al. (2024)	×	×	✓	✓	×	3	×
Peng et al. (2021)	×	×	×	×	×	7	×
Zheng et al. (2018)	✓	×	×	✓	×	7	×
Qi et al. (2022)	×	×	✓	×	✓	7	✓
Gu et al. (2022)	×	×	✓	✓	×	6	×
Sethi et al. (2022)	×	×	×	×	×	3	×
Proposed AIF	✓	✓	✓	✓	✓	7	✓

1.5 Contributions

The present study introduces a combined supervised and unsupervised data-driven approach utilizing advanced algorithms for the purpose of anomaly identification in AMI. The principal findings of our study include;

1. An innovative AI-based AIF is proposed integrated with a denoising autoencoder for noise robust feature transformation and a fine-tuned MLP classifier. This results in a highly effective AIF that demonstrates improved efficiency compared to state-of-the-art anomaly detection systems such as those presented in (Mishra and Das 2025; Qi et al. 2022; Peng et al. 2021).
2. The statistical association between information sets are assessed to articulate the results of preprocessing and verified through the utilization of the pearson correlation coefficient (PCC) and the maximum information coefficient (MIC) (Zheng et al. 2018). The preprocessing of the information set involves the removal of NaN (not a number), missing, and zero values in order to improve the statistical correlation of the information in the dataset.
3. Innovative domain specific synthetic feature sets are engineered, adopted from a modern smart meter data (Ajiboye et al. 2024) that improve sensitivity to abnormal patterns. Authors have created twelve (12) synthetic characteristics feature sets in this innovative approach. This method effectively mitigates the occurrence of false alarms that may arise from low energy usage among typical users during specific periods.
4. The authors have evaluated multiple sophisticated AI/ML-based models to detect anomalies and cyberattacks in AMI integrated with SG. The proposed AIF offers robust detection capabilities with improved generalization and reliability for multiple types of FDI attacks, including a mixed attack class that simulates complex real-world cyberattack scenarios.
5. Unlike (Mishra and Das 2025; Khan et al. 2025; Avila et al. 2018; Buzau et al. 2018) authors have utilized MLP for classification and denoising autoencoder for feature transformation to increase accuracy of detection mechanism and reduce false alarms.
6. A comprehensive evaluation has been carried out for results analysis of the implemented AI/ML based AIF for FDIs detection in AMI integrated with SG, that includes ablation studies with comparison against state-of-the-art approaches adopted form (Sharma and Tiwari 2024; Moriano et al. 2024; Hu et al. 2023; Jithish et al. 2023).

The results of the evaluation are unequivocally established the viability of the suggested approach for the purpose of detecting multiple FDIs in AMI. Moreover, our findings demonstrate that the suggested approach can uphold a commendable level of recognition efficacy despite a reduction in its detection using training data period from less than 55 days.

The rest of this article is structured as follows: In Section II, we provide a concise outline of the AMI architecture and investigation of seven different forms of FDI attacks, synthetic feature extraction, and design characteristics of autoencoder (AE), MLP, and support vector machine (SVM). Section III outlines the design and architecture of the AIF, encompassing with five phases of proposed AIF. An ample assessment of the AIF's performance is expounded in Section IV, covering key aspects like details of information sets, results and ablation study. The performance comparisons with existing techniques are given in Section V. Finally, in Section VI, we conclude the manuscript with a comprehensive conclusion with potential avenues for future research aspects.

2 System design and preliminaries

This section begins with an overview of the technological architecture of AMI, followed by a description of FDI attacks that can be used to simulate potential security threat to the power infrastructures. Furthermore, synthetic feature creation and AI/ML models details are given in this section.

2.1 AMI network design

Figure 4 illustrates the architectural layout of the AMI. As part of the power network design, smart meters are installed in both residential and commercial premises to monitor and record electricity consumption data. The detailed consumption data collected through these IoT-enabled devices. That can support various applications, such as smart energy management and demand side response programs. Energy data gathered from a group of nearby smart meters is transmitted to a central aggregation point through a neighborhood-area network. The wide-area network (WAN) then connects these neighborhood-area networks to power control centers. This layout forms the backbone of the AMI communication system.

The control center is accountable for managing smart meter data and overseeing the automated distribution of power flow in distribution networks. It is assumed that a neighborhood-area network may include a defined number of observer meters. Each observer meter records the total energy consumption data of a group of energy consumers. Further-

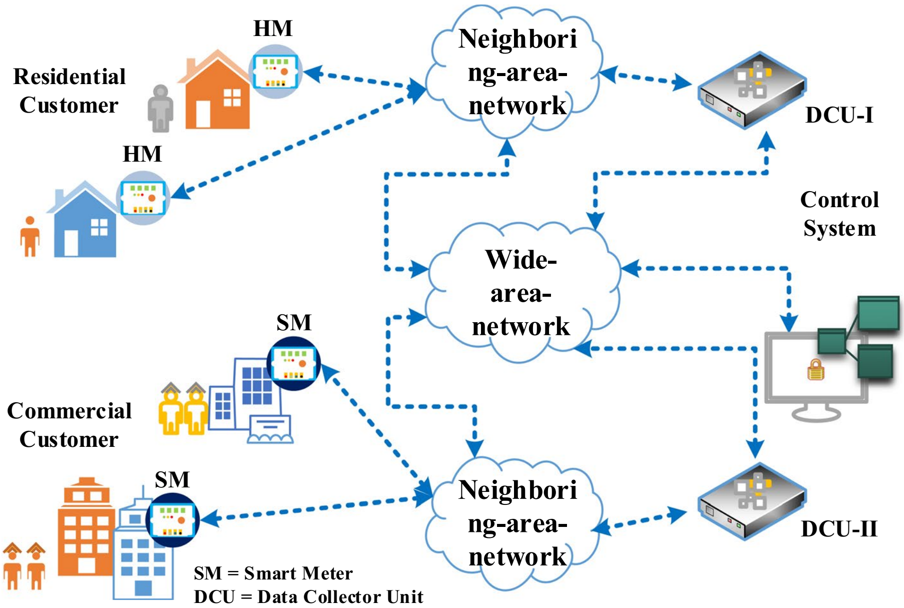


Fig. 4 Comprehensive layout and architectural design of the AMI data network

more, an observer meter can be positioned near the data concentrator to track the combined electricity usage of all customers within that network. Compared to standard smart meters, observer meters are generally more secure and resistant to tampering attempts aimed at falsifying energy consumption data (Sethi et al. 2022). As a result, the measurements recorded by observer meters are considered a reliable reference for the total consumption in a specific area. The communication networks in AMI, when integrated with smart grid systems may encounter several challenges. That include cybersecurity risks, secure data handling, and privacy concerns, due to the extensive interconnectivity and continuous data exchange (Chen et al. 2025; Nadeem and Hongsong 2025).

2.2 FDI attack types

Various types of FDI attacks can be used to simulate the malicious behavior of intruders for altering energy demand patterns. In this research study, six FDI attack types (FDI1â€"FDI6), as identified in existing literature (Peng et al. 2021), are analyzed and investigated. Additionally, the impact of a seventh variant (FDI7) is examined to assess the performance of the proposed AIF. Mathematical description of these FDI attacks are provided in Table 2.

The authentic and altered power usage values, denoted as $c_\mathcal{T}$ and $\bar{c}_\mathcal{T}$, accordingly, within a given period τ. The notation $\max(c)$ refers to the maximum value of the variable c, while $\bar{c}$ indicates the average daily power consumption recorded by the smart meter before any unauthorized manipulation.

As shown in Table 2, an FDI1 reduces all smart meter readings during a specified interval τ by applying a randomly generated scaling factor between 0.2 and 0.8. In FDI2 attacks, consumption data is reported only when the meter reading exceeds a randomly chosen threshold that is less than the maximum recorded value, $\max(c)$. FDI3 attacks involve subtracting a randomly selected value η. which is less than $\max(c)$ from the original collected readings. In FDI4 breaches, the smart meter data within a defined time window (τ_1, τ_2) is entirely erased and replace with 0.

The FDI5 attack alters the smart meter readings within a defined time interval τ by applying a randomly selected scaling factor $\kappa_\mathcal{T}$. That is ranges between 0.2 and 0.8. This approach helps reduce false positives and enhances the scalability and accuracy of training the AIF for detecting both FDI5 and FDI1 attacks. In FDI6 breach, the energy recorded values within the time window τ are replaced with the average daily consumption multiplied by a scal-

Table 2 Seven different forms of FDI breach types

Breach form	Alterations
1	$\bar{c}_\mathcal{T} = \kappa c_\mathcal{T}, 0.2 < \kappa < 0.8$
2	$\bar{c}_\mathcal{T} = \begin{cases} c_\mathcal{T} \text{ if } c_\mathcal{T} \leq \lambda \\ \lambda \text{ if } c_\mathcal{T} > \lambda \\ \text{where } \lambda < \max(c) \end{cases}$
3	$\bar{c}_\mathcal{T} = \max(c_\mathcal{T} - \eta, 0), \quad \text{where } \eta < \max(c)$
4	$\bar{c}_\mathcal{T} = f(\tau), \quad \text{where } f(\tau) = \begin{cases} 0 \text{ if } \tau_1 < \tau < \tau_2 \\ 1 \text{ otherwise} \end{cases}$
5	$\bar{c}_\mathcal{T} = \kappa_\mathcal{T} c_\mathcal{T}, 0.2 < \kappa_\mathcal{T} < 0.8$
6	$\bar{c}_\mathcal{T} = \kappa_\mathcal{T} \bar{c}, 0.2 < \kappa_\mathcal{T} < 0.8$
7	$\bar{c}_\mathcal{T} = \frac{1}{n}\sum_{j=1}^{n} c_{\mathcal{T},ij}, \quad \text{for each } i \in \{1, 2, ..., m\}$

ing factor κ_τ, also randomly selected within the range of 0.2 to 0.8 to mimic the real-time altering scenarios.

In FDI7, $c_\tau \in \mathbb{R}^{m \times n}$, where *m* is the number of rows representing the number of recorded days and *n* represents the number of columns, the features collected for each day or iteration in the collected dataset. Fig. 5 presents an example of a one-day user's energy consumption profile along with the corresponding modified profiles generated by applying six different types of FDI attacks. It is important to note that an individual attempting to falsify the electricity data may alter smart meter data by employing a combination of multiple FDI attack types.

2.3 Synthetic features for model training

Twelve (12) features were created from the SM energy consumption data using both statistical and electrical measurements. Creation of synthetic features process involves calculating several statistical and electrical features that characterize the dataset. Each feature function along with its corresponding mathematical model are,

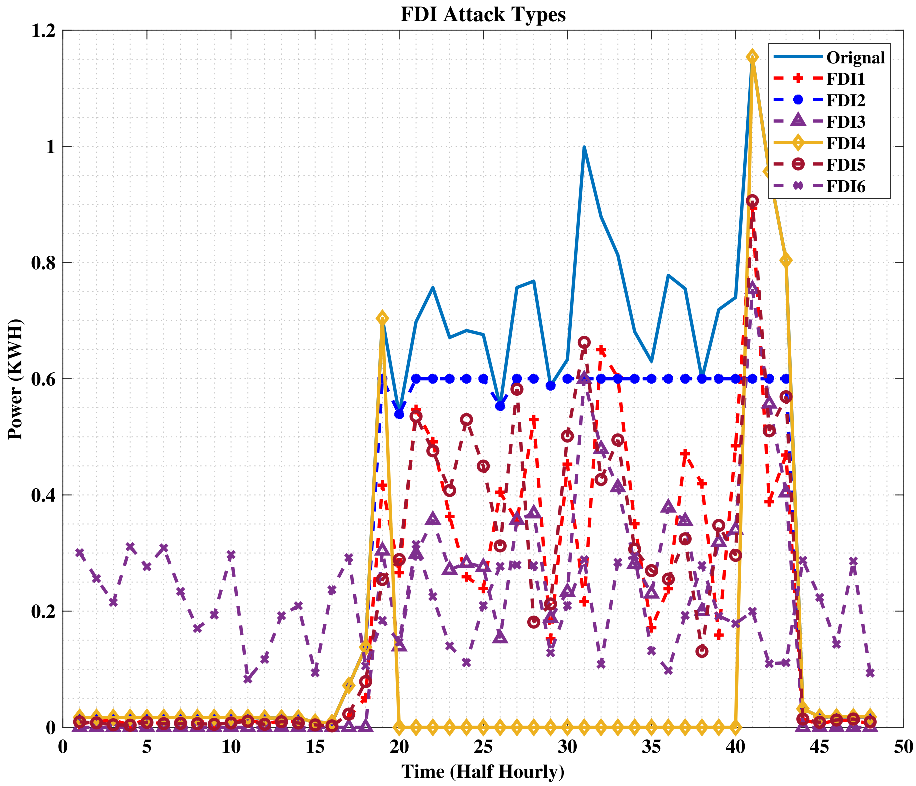


Fig. 5 One-day consumer profile with six different forms of FDI breaches

1. Minimum Energy Consumption: The minimum value of each entry is computed by,

$$\text{feature\_min}_i = \min_j(x_{ij}) \tag{1}$$

2. Maximum Energy Consumption: The maximum value of each entry is computed by,

$$\text{feature\_max}_i = \max_j(x_{ij}) \tag{2}$$

3. Mean of Energy Consumption: The mean value of each entry is computed by,

$$\text{feature\_mean}_i = \frac{1}{n}\sum_j x_{ij} \tag{3}$$

where n is the number of values in each entry.

4. Standard Deviation of Energy Consumption: The standard deviation of each entry is computed by,

$$\text{feature\_std}_i = \sqrt{\frac{1}{n}\sum_j (x_{ij} - \text{feature\_mean}_i)^2} \tag{4}$$

5. Skewness of Energy Consumption: The skewness of each entry is computed using the scipy.stats.skew function by,

$$\text{feature\_skew}_i = \frac{1}{n}\sum_j \left(\frac{x_{ij} - \text{feature\_mean}_i}{\text{feature\_std}_i}\right)^3 \tag{5}$$

6. Mean Absolute Deviation: The mean absolute deviation (MAD) of each entry is computed by,

$$\text{feature\_MAD}_i = \frac{1}{n}\sum_j |x_{ij} - \text{feature\_mean}_i| \tag{6}$$

7. Peak to Average value of Energy Consumption: The peak-to-average ratio of each entry is computed by,

$$\text{feature\_peak\_to\_avg}_i = \frac{\text{feature\_max}_i}{\text{feature\_mean}_i} \tag{7}$$

8. Peak to Peak value of Energy Consumption: The peak-to-peak value of each entry is computed by,

$$\text{feature\_peak\_to\_peak}_i = \text{feature\_min}_i + \text{feature\_max}_i \tag{8}$$

9. Current Measurement: The current (I) for each entry is computed using the formula by,

$$I_i = \frac{1000 \times KWH_{D_i}}{V_i \times 24 \times \cos\phi} \tag{9}$$

whereas, V_i is low voltage distribution voltages of $i-$th energy consumer.

10. Active Power Measurement: The active power (P) for each entry is computed by,

$$P_i = V_i \times I_i \times \cos\phi \tag{10}$$

11. Reactive Power Measurement: The reactive power (Q) for each entry is computed by,

$$Q_i = V_i \times I_i \times \sin\phi \tag{11}$$

12. Load Resistance: The resistance (R) for each entry is computed by,

$$R_i = \frac{V_i}{I_i} \tag{12}$$

These computed features collectively form the synthesized dataset, which captures various statistical and electrical characteristics of the input data. Assumptions and constraints utilized for creation of synthetic dataset are listed in Table 3. The choice of these twelve features is motivated by the capabilities of modern smart meters, which are increasingly equipped with built-in functionality to record not only basic energy usage but also detailed statistical, temporal, and behavioral parameters. Recognizing this, authors have designed synthetic features that simulate such built-in metrics. Additionally, the synthetic features set creation is inspired by real-world smart metering capabilities, guaranteeing that the feature engineering process aligns with practical deployment scenarios. These features enrich the input dataset with domain-relevant indicators, thereby enhancing the AIF's sensitivity to anomalous behaviors under various types of FDI attacks.

2.4 Autoencoder

The AE is a type of neural network algorithm design for unsupervised learning. It consists of three main layers; input layer, a bottleneck layer, and the output layer. As shown in Fig. 6,

Table 3 Variables and assumptions for creation of synthesized features

Variable	Equation	Description
1	$KWH_{D_i} = \sum_j x_{ij}$	The total kilowatt-hours per day (KWH_D) for each entry is calculated by summing the values along each row of the dataset
2	$V_i \sim \text{Uniform}(195, 209)$	Distributed voltage assumption
3	$\cos\phi = 0.88$	In residential and commercial power distributions, typically power factor ranges between 0.85 to 0.95. The selected value is reflecting a realistic operating condition observed in AMI integrated with smart grids.
4	$\sin\phi = \sin(\cos^{-1}(0.88))$	Q factor calculation

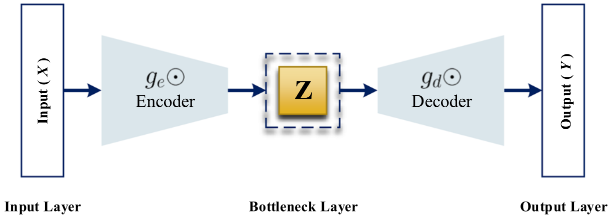


Fig. 6 Architecture of denoising autoencoder

the main function of an AE is to take the original input data and turn it into a smaller, simpler version called a code, while also trying to recreate the original data from this smaller version with as few errors as possible. This process comprises of two fundamental steps encoding and decoding. During the encoding phase, the input data is transformed into a low-dimensional latent space. This serves as a compressed version of the original data. In the decoding phase, the network attempts to reconstruct the original data from this latent representation (Berahmand et al. 2024). This ensures that the reconstruction closely resembles the input data. During encoding input data will be transformed into code which is representation of low-dimension data.

$$Z = g_e \odot (X) = \sigma_{Mish}(W_e X + b_e) \tag{13}$$

In (13) the input feature vectors are represented as $X = \{x_1, x_2, x_3, \ldots, x_t\}$, while $Z = \{z_1, z_2, z_3, \ldots, z_m\}$ represents the low-dimensional encoded representation. In the second step it performs decoding, the decoder takes the compressed code from the bottleneck layer and reconstructs it back into a representation that approximates the original input data. This reconstructed data is then compared with the original input to measure the difference. This allows the model to minimize reconstruction errors and improve the accuracy of the mapping process.

$$Y = g_d \odot (Z) = \sigma_{Mish}(W_d Z + b_d) \tag{14}$$

In (14), $Y = \{y_1, y_2, y_3, \ldots, y_t\}$ describes the output layer vector. The weights connecting the input to the hidden layer and the hidden layer to the output layer are denoted by W_e and W_d, respectively. The activation functions of the encoding and decoding layers are represented by $g_e\odot$ and $g_d\odot$, respectively. These establish the relationship between the encoded representation and the input data. Here in this research study, the *Mish* is employed as the activation function instead of the standard *ReLU*. This is a smooth and non-monotonic activation function defined as,

$$\sigma_{Mish}(x) = x \cdot \tanh(\ln(1 + e^x)) \tag{15}$$

The choice of *Mish* allows for better smoothness and self-regularization (Cai et al. 2022). This provides better overcoming the limitations of *ReLU* in such scenarios. Once the parameters of the encoder and decoder are configured, the model minimizes the difference between the original input data and the reconstructed output data. This is achieved by optimizing the mean square error (*MSE*) as given in (16). This evaluation confirms the accurate reconstruction through the iterative training process. Furthermore, *MSE* used in the AE reconstruction loss is central to distinguishing between normal and anomalous behavior. Since the autoencoder is trained only on normal data for feature transformations, it learns to reconstruct it with minimal error and reduced false positives.

$$MSE = \frac{1}{N}\sum_{i=1}^{N}(X_i - Y_i)^2 \tag{16}$$

here, N represents the total number of samples/records in the information set. It has been discussed a AE where the input and output are identical. While effective for reconstruction, this approach makes the AE highly sensitive to the training data. This limits its ability to generalize to unseen data. To address this limitation, a denoising AE introduces a slight modification by intentionally corrupting the input data with random noise before passing it through the network. This process compels the AE to reconstruct the original uncorrupted data from the noisy input. This makes it more robust and less sensitive to the specific characteristics of the training data. This enhancement improves the AE's generalization ability and significantly strengthens its capacity for effective feature extraction from noisy information sets. Hyperparameters are listed in Table 4. The selection of a 4 layer structure with layer sizes of [24, 24, 12, 12] is determined through preliminary experiments to trade-off between reconstruction accuracy and computational efficiency. A learning rate of 0.01 is selected with the Adam optimizer and a *StepLR* scheduler to ensure stable and consistent convergence over the 40 training epochs (Ni et al. 2024; Ravula et al. 2024). The batch size of 32 was selected to optimize memory usage and model generalization.

2.5 Multilayer perceptron

MLP is a class of feed forward neural network commonly employed for tasks related to regression and classification (El-Ghaish et al. 2024; Xia et al. 2022). The MLP comprises of multiple layers and nodes, including an input layer, hidden layers, and an output layer.

Table 4 Denoising autoencoder hyperparameters including network architecture, learning configuration, and optimization settings

Hyperparameter	Value	Hyperparameter	Value
Number of hidden layers	4	Optimizer	*Adam*
Layer sizes	[24, 24, 12, 12]	Learning rate scheduler	StepLR
Activation function	*Mish*	Learning rate	0.01
bottleneck layer	12	Number of epochs after which the learning rate is decayed	20
Loss function to minimize	*MSE*	Number of input features	Shape of training data
Multiplicative factor for learning rate decay	1.0	Batch size	32
Number of training epochs	40	Decay factor for Exponential Moving Average	0.70

In this implementation, the network architecture comprises four hidden layers with variable numbers of units are 100, 50, 50, and 20 respectively in each layer. The architecture of MLP is depicted in Fig. 7. The choice of the number of layers and units is often determined empirically and may depend on the complexity of the problem at hand. The activation function utilized in the hidden layers is the logistic *ReLU* function. With utilization the *ReLU* activation function, the output of each neuron in the hidden layers of MLP is computed as (17).

$$h_i^{(l)} = \max(0, \sum_j W_{ij}^{(l)} h_j^{(l-1)} + b_i^{(l)}) \tag{17}$$

where $W_{ij}^{(l)}$ and $b_i^{(l)}$ are the weights and biases for layer l, and $h_j^{(l-1)}$ is the output of the previous layer. In optimized MLP final layer uses the softmax function to produce probabilities for classification as given in (18).

$$y_k = \frac{\exp(x_k)}{\sum_{j=1}^{C} \exp(x_j)} \tag{18}$$

where $x_k = \sum_i W_{ki}^{(l)} h_i^{(l-1)} + b_k^{(l)}$ is the weighted sum for class k. The model is trained by minimizing the cross-entropy loss, the expression in given in (19)

$$\mathcal{L} = -\frac{1}{N} \sum_{i=1}^{N} \sum_{c=1}^{C} y_{i,c} \log(\hat{y}_{i,c}) \tag{19}$$

where $y_{i,c}$ is the true label and $\hat{y}_{i,c}$ is the predicted probability for class c. The weights and biases are optimized using gradient-based method i,e. Adam. The ReLU activation introduces non-linearity, that enables the MLP to model complex relationships in the data and helps in reduction noise propagation and overfitting, contributing to reducing false positives and enhancement in accuracy. The characteristics and hyperparameter of MLP are listed in Table 5. A 4-layer architecture with layer sizes of [100, 50, 50, 2] after hyperparameter tuning based on classification performance across multiple FDI types was selected. The *ReLU* activation function was adopted due to its effectiveness in mitigating vanishing gradient problems in deep networks. The learning rate of 0.0001 with the *Adam* optimizer ensured fast convergence and good generalization. Regularization parameters such as alpha = 0.0001 and validation fraction = 0.1 were used to reduce overfitting and improve robust-

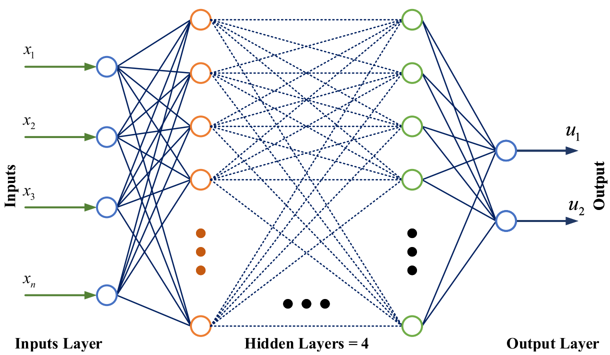


Fig. 7 Multilayer perceptron neural network architecture

Table 5 Hyperparameters used for training the MLP model, including network architecture, learning configuration, and optimization settings

Hyperparameter	Value	Hyperparameter	Value
Number of hidden layers	4	Shuffle	*True*
Hidden layer sizes	[100, 50, 50, 20]	Batch size	Auto
Activation function	*ReLU*	Learning rate unit	0.0001
Alpha	0.0001	Solver	Adam
Validation fraction	0.1	Tolerance for optimization	$1e^{-4}$
β_1	0.9	Number of iterations with no change	10
β_2	0.99	validation fraction	0.1
Epsilon	1×10^{-08}	Inverse scaling learning rate	0.5
Maximum iterations	200	Momentum for gradient descent update	0.9
Max function	15000	Warm start	*False*

ness. Other parameters such as β_1, β_2, and maximum iterations were chosen following standard practices validated through sensitivity analysis.

Remark 1 The hyperparameters for both the AE and MLP were determined through a series of preliminary experiments using a manual tuning approach. That was based on classification performance and reconstruction loss. While a full grid search was not used due to computational constraints, authors have employed a heuristic trial-and-error process by evaluating combinations of model parameters. That includes number of layers, layer sizes, activation functions, learning rates, and batch sizes. The selected configuration was chosen based on achieving optimal balance between performance high accuracy, F1-Score, low MSE value, and computational efficiency.

2.6 Support vector machine mode (SVM)

SVM is a powerful supervised learning algorithm widely used for classification and regression tasks. The fundamental idea behind SVM is to find the optimal hyperplane that best separates the data points of different classes in a high-dimensional space. The optimal hyperplane is determined by maximizing the margin, which is the distance between the hyperplane and the nearest data points from each class, known as support vectors (Li 2024). Mathematically, for a binary classification problem, the decision function of an SVM can be represented as,

$$f(x) = \text{sign}(w \cdot x + b) \tag{20}$$

where w is the weight vector, x is the input vector, and b is the bias term. The function sign returns the sign of the expression $w \cdot x + b$, i.e., it outputs $+1$ if the expression is positive (indicating one class) and -1 if negative (indicating the other class). The optimization problem for finding the optimal hyperplane can be formulated as:

$$\min_{w,b} \frac{1}{2}\|w\|^2 \tag{21}$$

subject to the constraint:

$$y_i(w \cdot x_i + b) \geq 1, \quad \forall i \tag{22}$$

where y_i are the class labels of the input data points x_i. In cases where the data is not linearly separable, SVM can be extended using kernel functions to map the input data into a higher-dimensional space where a linear separation is possible. The kernel trick allows SVM to efficiently perform non-linear classification by computing the dot product of the data points in the transformed space without explicitly performing the transformation.

3 Design of the anomaly identification system

The proposed methodology for detecting FDI attacks in smart metering infrastructure is illustrated in the Fig. 8. The methodology is divided into five phases, each critical for processing, feature extraction, and classification. The details of each steps are as follows.

3.1 Phase I: data transformation and preprocessing

To begin with phase-I, the process starts data collection, where SMs data is gathered, capturing consumer IDs, kWh consumption, and timestamps. Moving to next step Data Transformation, the raw data undergoes restructuring into a structured format across 48 time intervals, associating IDs with their respective features. For instance, data is organized as ID(1 : 48). Following this, data preprocessing involves cleaning the dataset by eliminating missing values (NaNs) and zero entries to ensure accurate analysis and interpretation.

3.2 Phase II: attack vector creation and labeling

In Phase-II, attack vector creation and labeling, the process begins with step, creating attack vectors, where various types of FDI attacks are introduced into the dataset, categorized into seven distinct attack scenarios. Following this in next step: label assignment, each data instance is assigned a label (0) signifies normal data, while (1) indicates an attack. Subsequently, synthetic feature creation involves generating 13 additional synthetic features. These include statistical and electrical features such as Kwh_{min}, Kwh_{max}, Kwh_{mean},

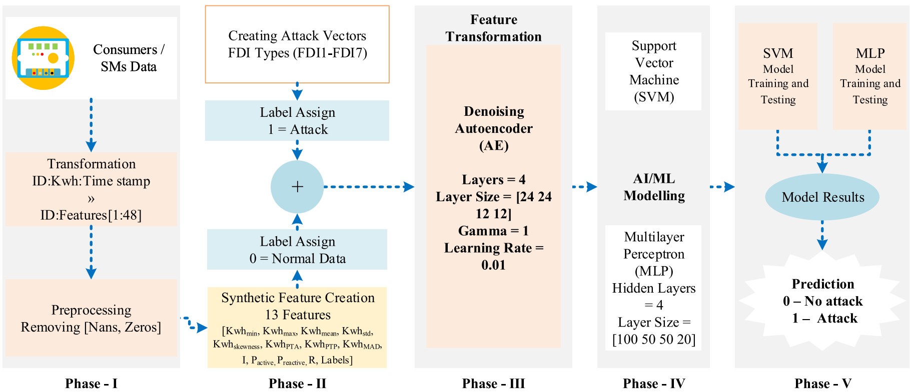


Fig. 8 Proposed layout architecture of anomaly identification system

Kwh_{std}, Kwh_{skewness}, $Kwh_{\text{P-T-A}}$, $Kwh_{\text{P-T-P}}$, Kwh_{MAD}, I (current), P_{active} (active power), P_{reactive} (reactive power), R (resistance), and labels. Additionally, a MIX type, referred to as the mixed type of attack, represents a composite scenario in which multiple FDI types among FDI1 - FDI7 are applied in combination or succession. This was included to assess the robustness and generalizability of the proposed AIF framework under conditions where multiple attack strategies may occur simultaneously or sequentially, as is plausible in real-world CPSs considering AMI integrated with smart grids.

3.3 Phase III: feature transformation using AE

In this phase, the focus was on transforming the extracted features from the AMI dateset into a more compact and meaningful representation to enhance the accuracy of anomaly detection. An AE, an unsupervised neural network model was employed for this purpose. The AE of four layers with a structured layer size of [24, 24, 12, 12], a gamma value of 1, and a learning rate of 0.01. This model works by encoding high-dimensional input data into a lower-dimensional representation while preserving essential patterns and structures. The primary objective of this transformation is to highlight hidden correlations and anomalies within the data that might not be easily detectable in its raw form. By reconstructing the input from its compressed representation, the AE identifies discrepancies, which can be useful in distinguishing attack instances from normal data. This feature transformation significantly improves the model's ability to detect FDI attacks by reducing noise and irrelevant information, making the subsequent classification process more efficient and accurate. This phase ensures that the dataset is optimally prepared for training the AI/ML models in the next phase.

Figure 9 shows the training and testing loss curves of the implemented AE with over 40 epochs. Initially, both the training loss and testing loss are high, however, with a rapid decrease in the first few epochs. This indicates the AE is quickly adapting to the data. By around the 5th epoch, both losses converge to low, stable values. This demonstrates the AE has effectively learned to reconstruct the input data without overfitting. The close alignment of the two curves also suggests good generalization performance.

3.4 Phase IV: AI/ML modeling

In this phase, the transformed feature set obtained from the AE were utilized to train and test machine learning models for FDI attack detection. Two widely used supervised AI/ML techniques, SVM and MLP were employed for classification. SVM is a powerful supervised learning algorithm used to find an optimal hyperplane that separates attack and non-attack instances. This ensures high generalization capability. On the other hand, MLP is a deep learning-based neural network with four layers and a layer size of [100, 50, 50, 2] were selected to train the model for to learns complex, non-linear relationships within the dataset to improve classification accuracy. The training process involves feeding labeled data into both models allowing them to recognize patterns, statistical anomalies, and deviations associated with cyberattacks. Once trained the models undergo testing on unseen data to evaluate their effectiveness in distinguishing between normal and attack instances. The dataset is randomly divided into 80% for training and 20% for testing.

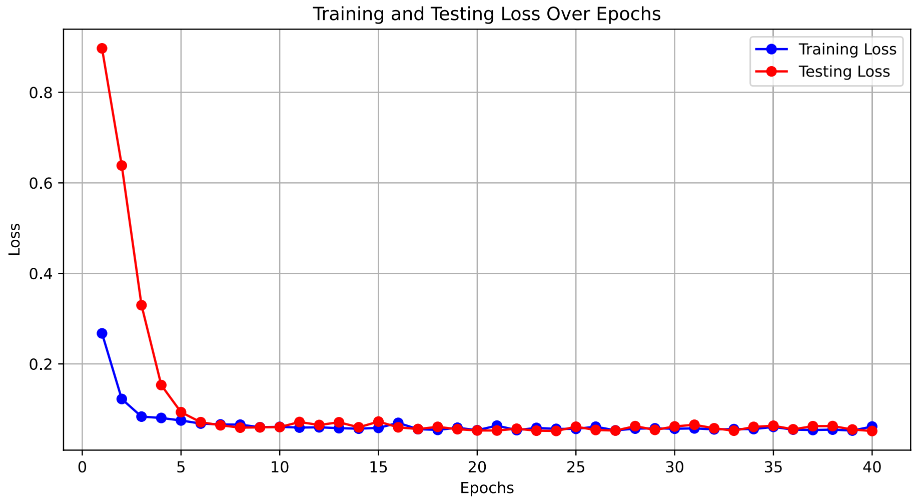


Fig. 9 Training and testing loss curve for validation of AE performance

3.5 Phase V: model results and prediction

The final phase of the proposed framework focuses on evaluating the trained machine learning models and generating predictions based on new, unseen data. The two models used in this study, SVM and MLP, undergo rigorous testing to determine their effectiveness in detecting FDI attacks in the AMI. The evaluation process involves assessing key performance metrics, which help measure how well the models differentiate between normal and attack instances. The trained models process the input data and classify each instance into one of two categories; “0" (no attack), representing normal operational data and “1" (attack) indicating a potential FDI attack. These predictions provide crucial insights into the security of the power grid, enabling utility providers to take proactive measures to mitigate cyber threats. By accurately identifying attack instances, the framework enhances grid reliability, prevents financial losses, and ensures secure energy distribution. The successful implementation of this phase demonstrates the viability of AI-driven anomaly detection systems in modern SG environments. This comprehensive methodology ensured the robust detection of FDI attacks in AMI by integrating advanced data processing, feature extraction, and AI/ML based supervised learning techniques.

4 Results and discussion

In this section information set and AIF results are given. Evaluating metrics such as, *MSE*, accuracy, F1-Score, recall, specificity, mean average precision (*MAP*), area under the curve (*AUC*) are calculated to validate the efficacy of proposed approach.

4.1 Information set

The information set used to evaluate the effectiveness of the AIF, consists of smart meter data from small and medium-sized enterprises (*SMEs*). This data originates from the Irish CER electricity smart metering initiative (Commission for Energy Regulation (CER) 2012), which has been widely referenced in studies focused on detecting electricity theft. Notably, this dataset is accessible after signing MOU with the company. It includes over 500 days of smart meter readings collected during 2009 and 2010 from more than 5, 000 residential users and 485 commercial entities, encompassing both small-scale and large-scale businesses. The dataset is considered clean and reliable, as all participants completed either preliminary or follow-up studies. Each smart meter records consumption at 30-minute intervals, yielding 48 readings per day. The segregation of dataset is illustrated in Fig. 10. For this research study, we utilized data from 485 commercial users over two consecutive 60-day periods: July 14 to August 13 and August 13 to September 12, 2009. Following a methodology similar to that outlined in (Zheng et al. 2018), the consumer population was divided into regions monitored by observer devices uniformly distributed across these areas. Within each region, a subset of consumers was selected to simulate cyberattacks, where 50% of their daily consumption profiles were subjected to various types of FDI attacks. For the FDIs in the dataset, the present work adopted the attack models from Qi et al. (2022); Peng et al. (2021). These models were selected due to their established relevance in FDI attack studies and their effectiveness in mimicking real-world cyberattacks in smart grid systems. To ensure a realistic evaluation, the attack strategies were carefully designed to manipulate energy consumption patterns while maintaining stealthiness against traditional detection mechanisms.

4.2 Evaluations results

The *MSE* results are given in Fig. 11a This illustrate the effectiveness of the AIF, which integrates a MLP with an AE to detect seven types of FDI attacks and a mix type of attacks. *MSE* serves as a dynamic metric for quantifying the disparity between predicted and actual values, where lower values signify superior model performance. These results highlight consistently low *MSE* values across all seven FDI attack types (FDI1 to FDI7) and the mixed attack category. This emphasized the framework's ability to reconstruct input data with minimal error. This indicates that the model effectively distinguishes between normal and anomalous data, achieving high detection accuracy. The testing accuracy of the AIF under various FDI attack scenarios are depicted in Fig. 11b The radar chart illustrates that the model achieves consistently high accuracy across multiple FDI attack types with performance exceeding 91%. The overall trend demonstrates the AIF robustness and reliability

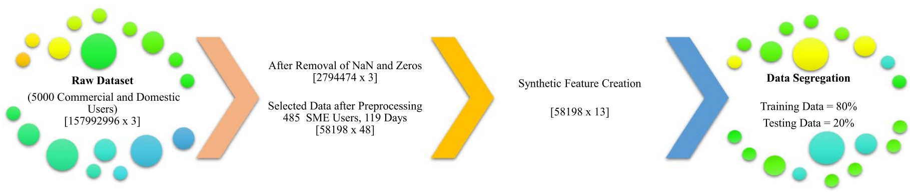


Fig. 10 Information set selection and segregation steps

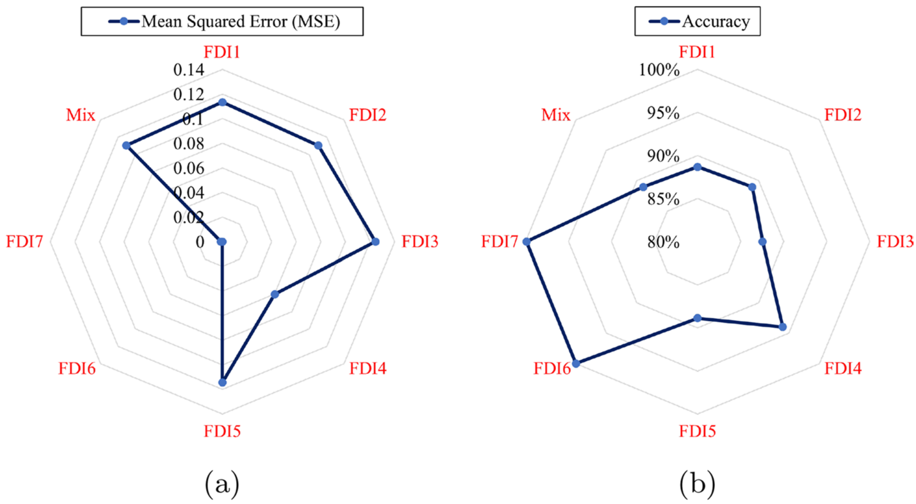


Fig. 11 **a** *MSE* results of model training by AE followed by MLP for detection of multiple FDIs, **b** testing accuracy under multiple FDIs using model trained by AE followed by MLP

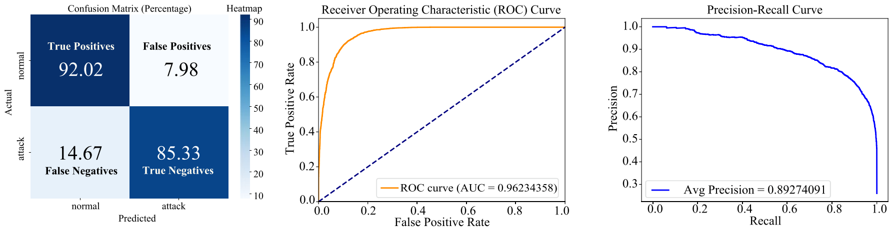
Table 6 AE followed by MLP model results for detection for FDIs

FDI type	F1-score (%)	Recall (%)	Specificity (%)	AUC-ROC (%)	MAP (%)
FDI1	88	85	92	96	91
FDI2	89	89	95	95	95
FDI3	88	88	93	94	95
FDI4	94	94	88	98	97
FDI5	90	90	93	97	92
FDI6	100	100	100	100	100
FDI7	100	100	100	100	100
Mix	89	89	91	95	96

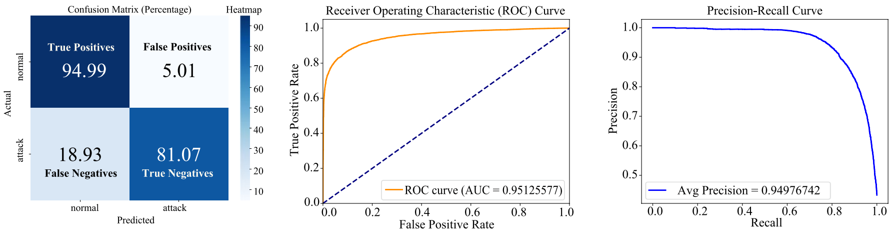
in detecting a wide range of FDI attacks with strong generalization to mixed and complex attack scenarios.

Table 6 listed the performance metrics of the AIF. The table reports critical evaluation metrics; F1-Score, Recall, Specificity, AUC-ROC, and MAP highlighting the framework's detection capability. The results indicate that the framework performs consistently well across most attack types. For individual attacks like FDI6 and FDI7, the model achieves perfect scores (100%) across all metrics, demonstrating its ability to detect these attacks with high precision and reliability. For other FDI types (FDI1â€"FDI5), the F1-Score, Recall, and Specificity values range between 85% and 95%, reflecting strong detection performance, although slightly lower compared to FDI6 and FDI7. The AUC-ROC values remain high across all cases, ranging from 94% to 100%. This emphasizing the model's ability to distinguish between normal and anomalous data effectively. Similarly, the MAP values, which measure precision across recall levels, are consistently above 91%, confirming the robustness of the framework. For the mix type of attacks, the model maintains high performance with an F1-Score of 89%, Recall of 89%, Specificity of 91%, and AUC-ROC and MAP values of 95% and 96%, respectively. These results demonstrate the framework's adaptability and generalization capabilities in handling complex attack scenarios. The table validates the effectiveness and robustness of the proposed model in detecting diverse FDI attacks in SG systems.

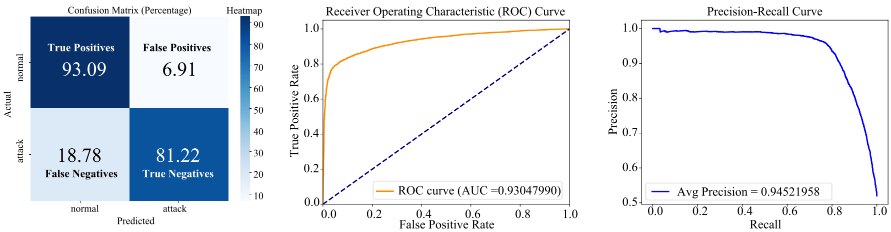
The results demonstrated in Fig. 12 and Fig. 13 comprehensively evaluate the effectiveness of the proposed AIF in detecting FDI attacks in AMI. The evaluation includes detailed analyzes of confusion matrices, receiver operating characteristic (ROC) curves, and preci-



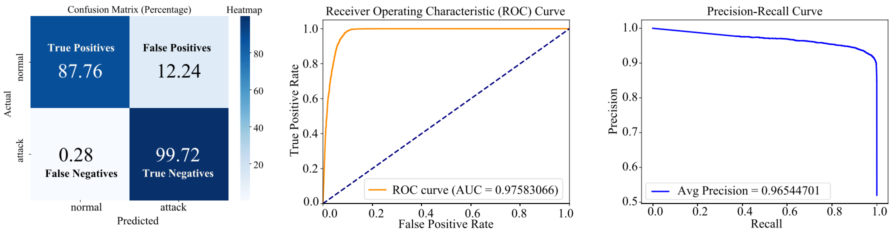
(a) Evaluation performance under FDI1; confusion matrix, ROC, and PR curves.



(b) Evaluation performance under FDI2; confusion matrix, ROC, and PR curves.

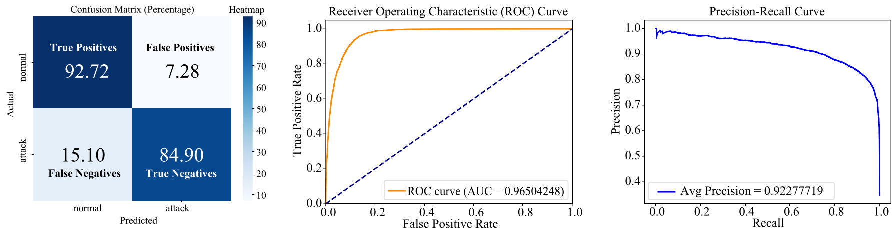


(c) Evaluation performance under FDI3; confusion matrix, ROC, and PR curves.

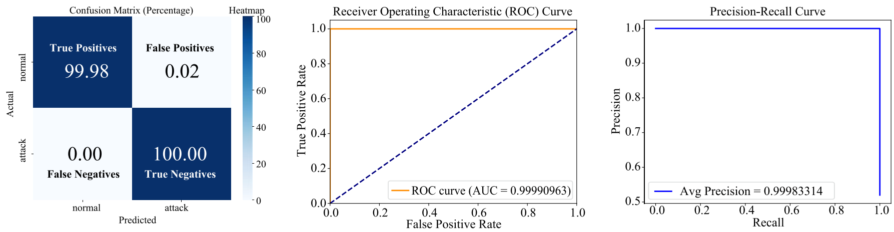


(d) Evaluation performance under FDI4; confusion matrix, ROC, and PR curves.

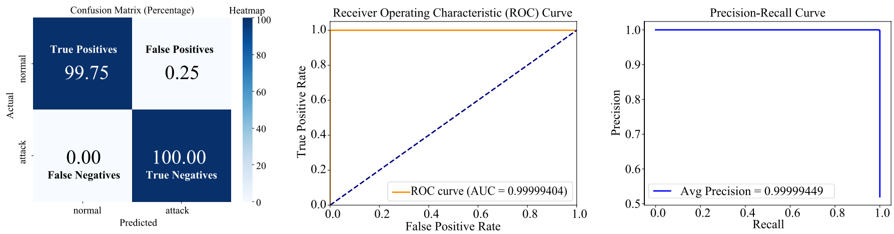
Fig. 12 Confusion matrices, AUC-ROC curves, and precision recall curves for AIF testing under FDI1-FDI4 attacks respectively



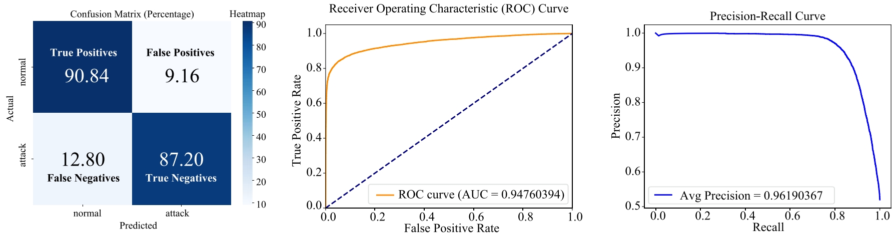
(a) Evaluation performance under FDI5; confusion matrix, ROC, and PR curves.



(b) Evaluation performance under FDI6; confusion matrix, ROC, and PR curves.



(c) Evaluation performance under FDI7; confusion matrix, ROC, and PR curves.



(d) Evaluation performance under MIX FDIs; confusion matrix, ROC, and PR curves.

Fig. 13 Confusion matrices, AUC-ROC curves, and precision recall curves for AIF testing under FDI5-FDI7 and MIX type attacks respectively

sion-recall (PR) curves for multiple attack types (FDI1 to FDI7) and a combination of mixed attacks. The results in the confusion matrices are normalized over rows.

The confusion matrices reveal the AIF's high classification accuracy with significant true positive (TP) and true negative (TN) rates across all attack types. When Ami subjected to FDI1, AIF achieved 92.02% for normal instances and 85.33% for attack instances, while FDI6 and FDI7 attain near-perfect scores with 100% accuracy. The false positive (FP) and false negative (FN) rates remain low, accentuating the model's reliability. For example, the FP rate for FDI4 is only 0.28%, and FDI6 and FDI7 exhibit virtually zero FP and FN rates. During mixed FDI attack scenarios, the framework maintains robust performance with a TP of 90.84% and a TN of 87.20%, demonstrating its adaptability to diverse and complex attack patterns.

The ROC curves and corresponding AUC values further validate the AIF's ability to distinguish between normal and anomalous data. All AUC values are exceptionally high, such as FDI1 (0.9623), FDI4 (0.9758), and near-perfect scores for FDI6 and FDI7 (0.9999 and 0.999994, respectively). Even for mixed attacks, the AUC is 0.9476, indicating strong generalization across multiple attack scenarios. The PR curves values confirm the framework's effectiveness in imbalanced datasets. High MAP values, such as FDI1 (0.8927), FDI4 (0.9654), and nearly perfect scores for FDI6 and FDI7 (0.9998 and 0.999994, respectively), demonstrate a strong balance between precision and recall. Even in mixed attack scenarios, the MAP value remains high at 0.9619, validating the model's precision and recall under complex conditions. The AIF exhibits high accuracy, robust performance under diverse attack scenarios, minimal false alarms, and strong generalization to multiple FDI attacks. These results establish the framework as a reliable and effective solution for detecting FDI attacks in AMI systems, emphasizing its potential for real-world deployment in SG integrated with AMIs.

4.3 Ablation study

Authors have performed ablation study which highlight the performance of the AE followed by an SVM (AE-SVM) framework in FDI attacks. The performance metrics in Table 7 listed the AE-SVM achieves reasonable results for most attack types with $F1$-Scores ranging from 68% for FDI2 to 100% for FDI6 and FDI7. However, its performance is inconsistent across different attack scenarios. While recall is high for FDI6 and FDI7 (100%), it is notably lower for FDI2 (69%), indicating challenges in detecting certain attack types. Specificity and AUC-ROC values are strong overall, but the MAP for FDI2 (67%) reveals room for improvement in balancing precision and recall. In mixed attack scenarios, the framework achieves an F1-Score and recall of 84%, which reflects moderate robustness but a decline compared to simpler attack scenarios.

The radar charts further emphasized these observations. The *MSE* results depicted in Fig. 14a reveal significant variability with lower values for FDI6 and FDI7 aligning with high accuracy but higher values for FDI1 and FDI2. This corresponds to lower detection accuracy. Similarly, the accuracy chart in Fig. 14b demonstrated a near-perfect detection

Table 7 AE followed by SVM model results for detection for FDIs

FDI type	F1-score (%)	Recall (%)	Specificity (%)	AUC-ROC (%)	MAP (%)
FDI1	86	81	92	94	83
FDI2	68	69	82	73	67
FDI3	83	83	83	90	92
FDI4	91	91	83	95	94
FDI5	86	81	92	94	83
FDI6	100	100	99	100	99
FDI7	100	100	99	100	99
Mix	84	84	92	91	92

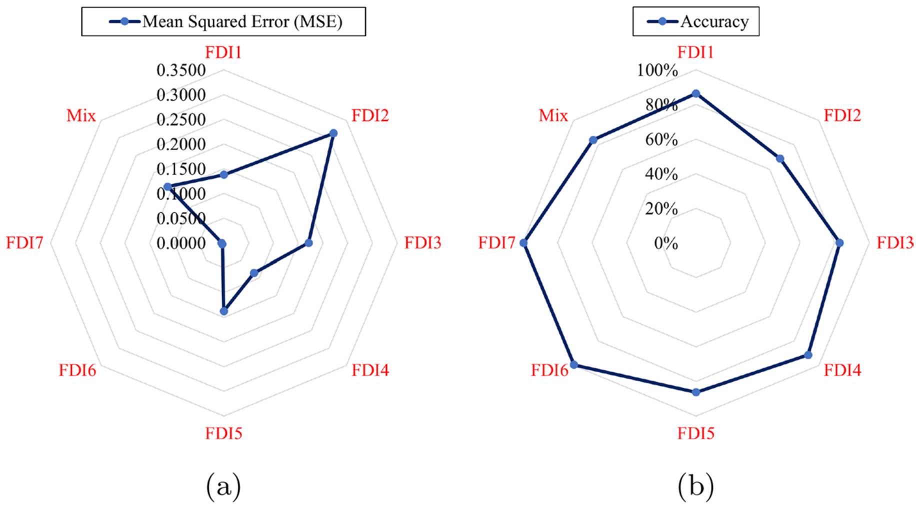


Fig. 14 **a** *MSE* results of model training by AE-SVM for detection of multiple FDIs, **b** testing accuracy under multiple FDIs using model trained by AE-SVM

for FDI6 and FDI7 but highlights poor performance for FDI2 and other complex attack scenarios. When comparing SVM to the MLP on feature transformed by an AE framework, the superiority of MLP becomes evident. The MLP model demonstrates consistently higher $F1$-Scores, recall, specificity, and AUC-ROC across all FDI types. AIF trained by AE followed by MLP (AE-MLP) achieved over 90% F1-Score and recall for FDI1 and FDI2, outperforming AE followed by SVM, which struggles with 68% and 69%, respectively. In mixed attack scenarios, AIF with AE-MLP also exhibits stronger robustness and precision-recall balance. Moreover, AE-MLP achieves lower *MSE* values across all attack types. This indicates better reconstruction of normal patterns and improved anomaly detection capabilities. The adaptability of MLP to capture complex patterns in data enables it to handle diverse and subtle FDI attack types more effectively than SVM, which is limited in its representation power. The higher MAP values achieved by AE-MLP further highlight its superior precision-recall trade-off, make it better suited for imbalanced data scenarios.

An additional ablation study was conducted using multiple advanced algorithms to detect the MIX type of FDI attacks, where four out of seven different FDI types were randomly selected to simulate realistic scenarios. The Table 8 summarizes the performance comparison among state-of-the-art advanced algorithms and the proposed approach AE encoded features for FDIs detection in AMI. These AI/ML based model are adopted from literature and extensively utilized in intrusion and anomalies detection in CPSs. Among the compared algorithms, random forest Moriano et al. (2024), adaptive boosting (AdaBoost) Hu et al. (2023), extreme gradient boosting (XGBoost) Sharma and Tiwari (2024), logistic regression Jithish et al. (2023), and the proposed AE-MLP, the proposed consistently achieves superior overall results when applied to Irish CER electricity smart metering dataset. The AE-MLP demonstrates the highest accuracy (88.32%), F1-Score (88.21%), recall (88.32%), AUC-ROC (94.25%), MAP (95.87%), and the lowest MSE of 0.1199. Although random forest slightly surpasses the proposed method in specificity, AE-MLP's balanced and robust performance across all metrics emphasizing its effectiveness and suitability for accurately

Table 8 Comparison among state-of-art advanced algorithms and proposed approach using AE-encoded features

Model	Accuracy (%)	F1-Score (%)	Recall (%)	Specificity (%)	AUC-ROC (%)	MAP (%)	MSE
Random Forest	87.89	87.88	87.89	93.00	94.07	95.52	0.1211
AdaBoost	80.43	80.39	80.43	86.75	87.92	91.13	0.1957
XGBoost	87.20	87.19	87.20	92.80	93.65	95.41	0.1280
Logistic Regression	78.32	78.31	78.32	82.31	85.76	88.67	0.2168
AE_MLP (Proposed)	88.32	88.21	88.32	90.31	94.25	95.87	0.1199

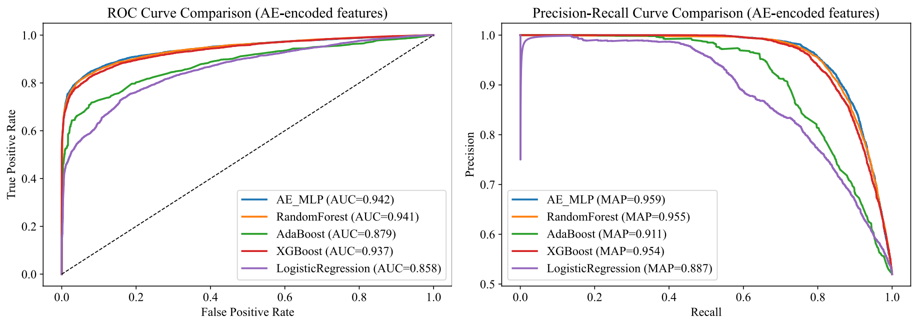


Fig. 15 Roc-AUC, and MAP combined curves of proposed and state-of-the-art algorithms

detecting anomalies caused by FDI attacks. Furthermore, the Fig. 15 presents ROC and precision-recall curves, where the AE-MLP model attains superior AUC of 0.942 and MAP of 0.959 values, that indicate robust FDI detection capabilities and effective handling of data imbalance. The Fig. 16 provides confusion matrices for all evaluated models, highlighting their classification performance. The AE-MLP model demonstrates balanced performance with an accuracy of 90.31% for correctly identifying normal instances and 85.88% accuracy for attack instances. Random forest and XGBoost models show slightly higher accuracy in normal instance detection. However, the AE-MLP model provides the best overall balance between false positives and false negatives. This combined analysis emphasizes that AE-MLP achieves consistent and reliable FDI detection. This makes the proposed highly suitable for securing AMI systems against FDI attacks.

5 Performance comparison with the existing results

5.1 Base-line methods for comparison analysis

Multiple base-line methods were selected for the performance comparison to validate the efficacy of the AIF. The brief description of each base-line methods are as follows,

1. Hierarchical density-based spatial clustering of applications with noise (HDBSCAN), a novel data driven technique utilized in (Mishra and Das 2025). HDBSCAN was used

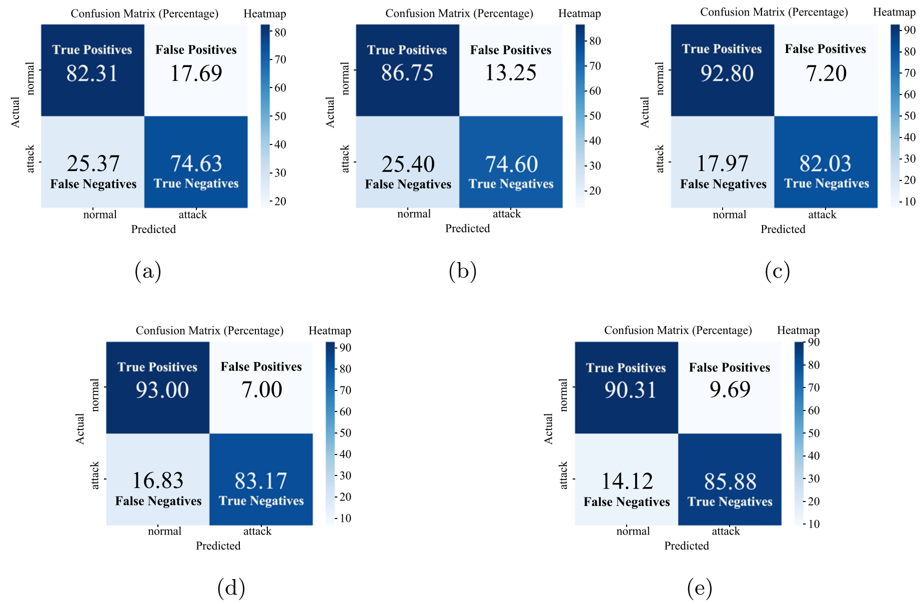


Fig. 16 Confusion matrices results using various algorithms; **a** logistic regression, **b** AdaBoost, **c** XG-Boost, **d** random forest, and **e** AE_MLP

to detect falsified data by identifying irregular consumption patterns from AMI data. It effectively detects outliers and noise points, that deviate from normal behavior.

2. Fuzzy C-means (FCM) is a state-of-the-art clustering technique to distinguish data points to belong to multiple clusters with varying degrees of membership. Authors in (Qi et al. 2022) utilized FCM with discrete wavelet transformation (DWT) to identify falsified data attacks in AMI integrated with SG.
3. Local outlier factor (LOF) is a traditional outlier detection approach in (Zheng et al. 2018; Peng et al. 2021) used to pinpoint electricity thieves based on users' electricity consumption profiles.
4. CLOF, a novel outlier detection technique was implemented in (Peng et al. 2021), which merges k-means clustering and LOF to recognize cases of electricity theft using users' energy consumption profiles.
5. Clustering by fast search and finding of density peaks (CFSFDP), a unique density-based clustering algorithm used in (Zhou et al. 2024) for identify arbitrarily shaped clusters and also utilized in (Zheng et al. 2017) for identifying electricity theft through users' energy consumption profiles.
6. Pearson correlation coefficient (PCC) and maximum information coefficient (MIC) are the Two frequently utilized techniques of correlation analysis (Peng et al. 2021) and (Zheng et al. 2018), were used to recognize instances of FDIs in cosnumption data. These techniques measure the correlation among the non-technical losses (NTL) and tempered consumer energy profiles. The NTL is calculated by subtracting the observer meter reading profile from the total of users' consumption profiles, which includes tempered profiles.

7. Combined data-driven approach (CDDA), a unique data-driven method that joins CFSFDP clustering and MIC-based correlation analysis for detecting falsifying electricity consumption data, as mentioned in (Zheng et al. 2018). The CDDA produces a fused rank for detection using the geometric mean (CDDA-G) and arithmetic mean (CDDA-A) of the ranks acquired by CFSFDP and MIC.

5.2 Performance comparison with baseline methods

Table 9 listed the results to validate the effectiveness of the AIF trained with an AE-MLP compared to the baseline methods across different forms of FDI attacks. The evaluation metric used is the AUC% and MAP%. These are widely accepted metrices for assessing classification performance of AI/ML based model. For FDI1, the proposed achieves an outstanding AUC of 96.23%, significantly overcoming all baseline methods including CDDA-G (76.90%), CDDA-A (72.50%), HDBSCAN (78.80%), FCM-DWT (94.4%), and PCC (80.80%). This demonstrates the robustness of the proposed framework in detecting this attack type. Similarly, for FDI2, the AIF scores 95.13%, whereas baseline methods, such as CFSFDP (58.40%), CDDA-G (64.30%), CDDA-A (65.70%), FCM-DWT (90.3%), and CLOF (41.30%), exhibit much lower performance, validating the superior adaptability of the proposed model. For FDI3, the proposed framework maintains its dominance with an AUC of 93.52%, while the FCM-DWT shows better response in this case. Notably, traditional outlier detection techniques like LOF (43.40%) and CLOF (44.70%) fall behind significantly proving that the AIF can identify intricate patterns in the data that other methods fail to capture.

A similar trend is observed for FDI4, where the AIF achieves an AUC of 97.58%, surpassing even the best-performing baseline method, CDDA-G (94.00%). However, the FCM-DWT AUC results are 0.52% better than the proposed. The proposed also excels in detecting FDI5, with an AUC of 96.50%, which is substantially higher than that of MIC (61.30%), PCC (40.60%) and state-of-the-art baseline methods. For FDI6, the AIF achieves a perfect AUC of 100%, far exceeding baseline methods with 9.6% to 64%, further demonstrating its superiority for complex attack scenarios. When dealing with mixed attack types, the AIF achieves an AUC of 94.76% validating its effectiveness in generalizing across multiple attack forms. In contrast, the baseline methods, including CDDA-A (73.20%) and MIC (61.70%), display inferior performance. Authors have achieved 99.99% AUC score in FDI7, which is not performed by existing baseline methodologies. These results validate the superiority of the proposed trained with AE-MLP. The model consistently achieves higher AUC values across all FDI types, outperforming baseline methods that rely on clustering, correlation analysis, and outlier detection. These findings emphasize the robustness, adaptability, and reliability of the proposed approach for anomaly detection in CPSs. The Fig. 17 illustrating the graphical comparison representation of AIF with baseline methods.

For statistically validation of the performance improvement of the proposed, the paired t-test comparing the AUC scores of the proposed method with baseline techniques across all seven FDI attack scenarios is conducted. The paired t-test is computed as (23) (Okoye and Hosseini 2024).

$$t = \frac{\bar{d}}{s_d/\sqrt{n}} \tag{23}$$

Table 9 Results comparison among proposed and baseline methods using AUC% as evaluating metric

Breach form	Proposed	CDDA-A Qi et al. (2022); Zheng et al. (2018)	CDD-G Qi et al. (2022); Zheng et al. (2018)	CFS-SDP Zheng et al. (2017)	PCC Peng et al. (2021)	MIC Zheng et al. (2018); Peng et al. (2021)	LOF Qi et al. (2022); Peng et al. (2021)	CLOF Qi et al. (2022); Peng et al. (2021)	HDB-SCAN Mishra and Das (2025)	FCM-DWT Qi et al. (2022)
FDI1	96.23	72.50	76.90	50.20	80.80	83.40	41.80	41.80	78.80	94.4
FDI2	95.13	65.70	64.30	58.40	59.90	59.40	45.40	41.30	-	90.3
FDI3	93.52	78.90	79.70	72.30	49.80	63.70	43.40	44.70	81.30	99.**6**
FDI4	97.58	93.50	94.00	92.00	28.60	74.40	59.70	61.90	81.00	98.1
FDI5	96.50	67.20	65.50	56.80	40.60	61.30	46.90	48.90	-	91.6
FDI6	100.00	65.20	65.10	75.30	36.00	41.20	42.00	40.90	84.70	90.4
FDI7	99.99	–	–	–	–	–	–	–	–	–
Mix	94.76	73.20	74.70	70.00	47.90	61.70	46.80	47.20	82.00	94.0

where $\bar{d}$ is the mean of the differences between paired observations (AUC of proposed method - AUC of baseline). s_d is the standard deviation of the differences and n is the number of paired samples, that is here in this case $n = 7$. The null hypothesis H_0 assumes that there is no significant difference in performance between the proposed and baseline methods. A p-value less than 0.05 rejects H_0 with 95% confidence. The paired t-test results for all baseline comparisons are summarized in Table 10.

The p-values for all comparisons are less than 0.05, that authenticates the improvements achieved by the proposed AIF is statistically significant across all considered FDI breaches in comparison with baseline methodologies.

Table 11 combined with the graphical representation in Fig. 18 demonstrated the superiority of the AIF trained with AE followed by MLP when evaluated using the MAP% metric. Across all breach forms, the AIF consistently outperforms baseline methods, showcasing its exceptional ability to accurately identify anomalies in CPSs. For FDI1, the AIF achieves a MAP of 90.94%, higher than the best-performing baseline method, PCC (68.60%). Other methods, such as CDDA-A (52.60%), HDBSCAN (68.00%), FCM-DWT (86.7%), and CLOF (17.10%) emphasizing the proposed model's ability to effectively capture the complexities of this attack type. In the case of FDI2, the AIF attains a MAP of 94.98%, significantly greater than baseline methods, with MIC achieving only 39.70% and PCC scoring 34.10%. This emphasizes the robustness of the proposed framework in detecting anomalies with high precision. Similarly, for FDI3, the AIF achieves 94.83%, while CDDA-A (54.40%) and CFSFDP (36.90%) show limited effectiveness. Although FCM-DWT is performing 2.97% better in term of MAP% as an evaluating metric when AMI is subjected to FDI3.

For FDI4, the proposed framework achieved a MAP of 96.54%, surpassing the CDDA-A by a wide margin. Traditional techniques such as LOF and PCC perform poorly, further validating the AIF's advanced anomaly detection capabilities. Similarly, for FDI5, the AIF scores 92.28%, compared to only 42.00% for MIC and 38.80% for CDDA-A, highlighting the model's superior precision. A MAP score of 100.00% is observed for FDI6, where all baseline methods perform inadequately, with MIC and CLOF achieving only 20.70% and 16.30%, respectively. This demonstrates the proposed model's unparalleled ability to detect complex and subtle anomalies. For the mixed attack scenario, the AIF achieves a MAP of 96.19%, significantly outperforming baseline methods such as CDDA-A (52.20%) and MIC (44.80%). These results emphasized the AIF's generalizability and effectiveness in handling diverse types of anomalies. These findings strongly validate the framework's potential for anomaly detection in complex CPSs. Furthermore, PCC and MIC are traditional correlation-based techniques, tend to fall short against FDI4 and FDI6. This is because these attacks are designed to create subtle but structured alterations that do not significantly affect correlation metrics. Similarly, LOF and CLOF struggle with FDI2 and FDI5, that maintain a degree of statistical normalcy. Due to unsupervised nature and reliance on local density, these methods may not effectively distinguish between legitimate but uncommon patterns and actual intrusions in AMIs. The CFSSDP, CDDA-A, and CDDA-G may misclassify data under complex FDI scenarios, such as MIX FDI attacks, due to overlapping feature distributions and insufficient discrimination power in high-dimensional space. These shortcomings emphasized the advantage of proposed AIF integrated with deep learning algorithms, that results in stronger generalization and adaptability to various FDI attack patterns.

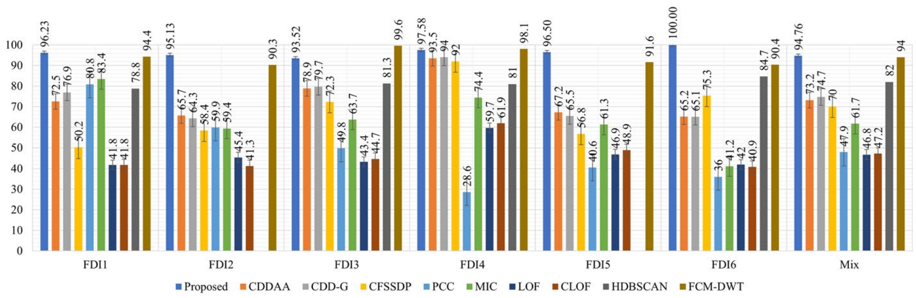


Fig. 17 Graphical representation of results comparison using AUC% as an evaluating metric

Table 10 Paired t-test results for performance evaluation

Baseline method	t-statistic	p-value
LOF	21.02	< 0.0001
CLOF	17.54	< 0.0001
PCC	6.84	0.0005
MIC	6.13	0.0009
CDDA-A	5.73	0.0012
CFSSDP	5.54	0.0015
CDD-G	5.22	0.0020
HDBSCAN	14.45	0.00013

The existing baseline methods CDDA-A, CDDA-G, LOF, CLOF, and HDBSCAN incline to rely on density-based assumptions and fixed clustering behaviors. That makes them less effective when dealing with sparse and refined modifications, as the behavior seen in FDI2-FDI7. These FDI types involve partial data suppression and localized distortions that blend in with normal variations. This make them difficult to distinguish using unsupervised and distance-based approaches. In distinction, the proposed AIF utilizes domain oriented synthetic features and enhanced learning outcome of the fine-tuned AE-MLP pipeline. This allows its ability to capture complex temporal and statistical deviations, that results in significantly better generalization and anomalies/cyberattack detection accuracy among multiple FDI types.

Remark 2 Across seven FDIs and one mixed attack type, the proposed AIF consistently outperforms FCM-DWT in six out of eight instances with higher AUC scores with prominent margins in stealthy and low-load conditions. This demonstrates enhanced robustness, generalization, and detection capability of the proposed AIF compared to state-of-the art anomaly detection methods.

5.3 Sensitivity analysis

For evaluating the robustness of the proposed AIF under realistic conditions where the proportion of attack data may vary, a sensitivity analysis is performed. The AIF was trained using MIX FDI type attacks datasets at three different intensity levels. That includes 20%, 30%, and 40%. As depicted in Fig. 19, the confusion matrices across these settings demon-

Table 11 Results comparison among proposed and baseline methods using MAP% as an evaluating metric

Breach form	Proposed	CDDA-A Qi et al. (2022); Zheng et al. (2018)	CDD-G Qi et al. (2022); Zheng et al. (2018)	CFS-SDP Zheng et al. (2017)	PCC Peng et al. (2021)	MIC Zheng et al. (2018); Peng et al. (2021)	LOF Qi et al. (2022); Peng et al. (2021)	CLOF Qi et al. (2022); Peng et al. (2021)	HDB-SCAN Mishra and Das (2025)	FCM-DWT Qi et al. (2022)
FDI1	90.94	52.60	57.40	27.80	68.60	59.90	16.90	17.10	68.00	76.9
FDI2	94.98	42.90	48.20	26.80	34.10	39.70	18.60	16.40	-	86.7
FDI3	94.83	54.40	58.90	36.90	24.40	42.80	17.30	20.80	91.50	97.8
FDI4	96.54	78.00	80.70	65.40	8.80	45.50	30.20	37.80	89.00	87.5
FDI5	92.28	38.80	45.80	25.90	14.60	42.00	21.60	22.70	-	68.1
FDI6	100.00	36.00	42.10	33.40	14.80	20.70	16.60	16.30	72.00	79.1
FDI7	99.99	-	-	-	-	-	-	-	-	-
Mix	96.19	52.20	56.30	39.60	32.60	44.80	20.40	22.20	78.00	84.2

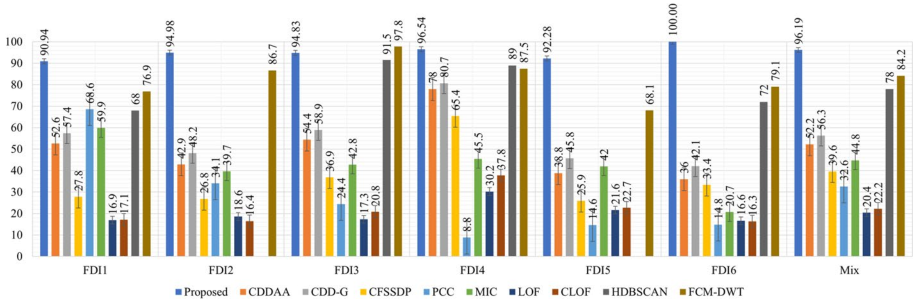
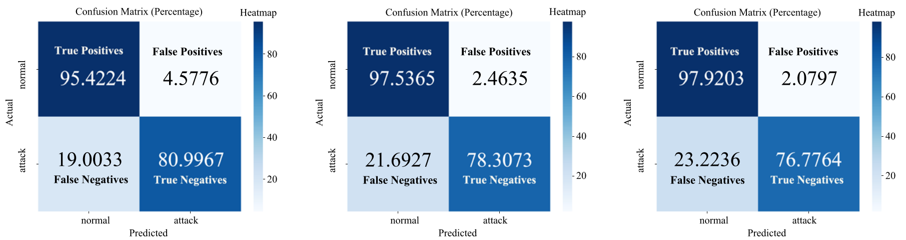


Fig. 18 Graphical representation of results comparison using MAP% as an evaluating metric

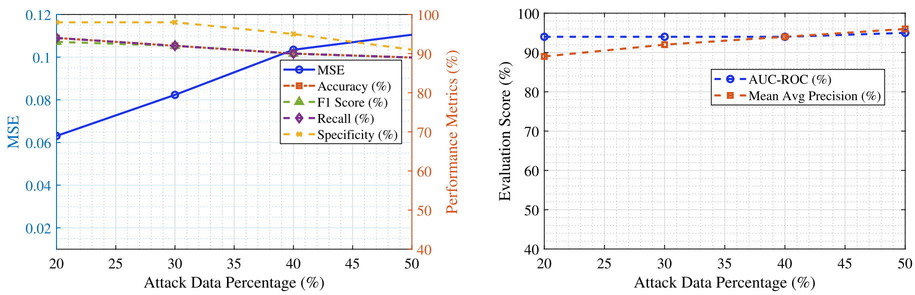


(a) Evaluation with 40% MIX attack data.

(b) Evaluation with 30% MIX attack data.

(c) Evaluation with 20% MIX attack data.

Fig. 19 Confusion matrices showing the robustness of the proposed AIF model with varying percentages of MIX-type attack data used in model training



(a)

(b)

Fig. 20 Evaluation metrics with varying percentages of MIX attack data for AIF model training

strated high true positive and true negative rates. This indicates the AIF's consistent classification performance despite variations in attack prevalence.

In Fig. 20, the variation in major performance indicators with respect to increasing MIX FDI attacks data percentages are presented. It is observed that the proposed AIF maintains high Accuracy (more than 90%), F1-Score, and Recall among all tested ratios. The MSE increases slightly with higher attack intensity, which is expected due to increased data complexity and realistic behavior. However, the both AUC-ROC and MAP remain high and stable. That confirms the AIF's generalization ability among different size of attach vector and datasets. These results authenticate the resilience of the proposed AIF in practical AMI integrated with SG scenarios where the ratio of malicious data may fluctuate dynamically.

6 Conclusion

Modern cyber physical systems are susceptible to multiple types of cyberattacks. In literature, multiple AI-based approaches have been investigated. These techniques primarily rely on supervised and unsupervised machine learning. However, supervised methods require a labeled dataset with legitimate and malicious load patterns, which limits their scalability and adaptability. While unsupervised approaches often struggle with accuracy and identifying complex patterns when subjected to multiple types of FDI scenarios. To overcome these challenges, the authors of this research study have developed an anomaly detection framework integrated with an AI/ML-based supervised MLP combined with an unsupervised AE to counteract FDI attacks in SG integrated with AMI. The proposed framework demonstrated superior performance over traditional detection techniques, exhibiting high accuracy, precision, and recall across various FDI scenarios. Authors have employed synthetic feature extraction, AE-based feature transformation, and AI-driven classification in this research. The combined framework significantly enhanced the detection capability while minimizing false alarms. Comparative analysis with baseline models confirmed its effectiveness, with the proposed approach achieving the highest AUC% and MAP% scores across multiple attack types. Future research should be focus on enhancing the generalizability and adaptability of AI-driven anomaly detection models to address evolving cyber threats such as replay attacks, coordinated DoS and FDIs in SGs. The integrating federated learning techniques can be investigated to enhance security by enabling decentralized attack detection without exposing sensitive consumer data. These advancements will contribute to a more resilient, intelligent, and secure power distribution network

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Author contributions All authors contributed equally to this work.

Data availability No datasets were generated or analysed during the current study.

Declarations

Competing interests All authors confirms that they do not possess any identifiable conflicting financial interests or personal affiliations that might have influenced the research presented in this manuscript.

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